Koneru Lakshmaiah Education Foundation
(Deemed to be University estd. u/s. 3 of the UGC Act, 1956)
Off-Campus: Bachupally-Gandimaisamma Road, Bowrampet, Hyderabad, Telangana - 500 043.
Phone No: 7815926816, www.klh.edu.in

PROMPTING VS RAG VS FINE-TUNING FOR CUSTOMER SUPPORT AUTOMATION

LITERATURE SURVEY

2520030069-G. Hemaswi

2520030091-K. Rupesh

Department: CSE

College: Koneru Lakshmaiah Education Foundation

Academic Year: 2026–2027

1. Introduction

Customer support teams handle large volumes of repetitive queries about orders, refunds, accounts and troubleshooting. Large language models (LLMs) can automate many of these responses, but a general-purpose model does not know a company's products, policies or preferred tone, and it may produce inaccurate or fabricated answers. Three techniques are commonly used to adapt an LLM to this setting: prompt engineering, Retrieval-Augmented Generation (RAG) and fine-tuning. They change different things: prompting changes the instructions, RAG changes the information supplied at answer time, and fine-tuning changes the model's learned behavior.

This literature survey reviews research relevant to choosing among these techniques for customer support automation. It focuses on studies that compare RAG and fine-tuning, on work that combines them, and on customer-support applications of RAG. The review also identifies a research gap: a controlled evaluation that compares prompting, RAG, fine-tuning and their hybrids on the same support tasks, while jointly measuring accuracy, hallucination, tone, latency, cost and adaptability to policy changes.

2. Key Concepts

2.1 Prompt engineering

Prompt engineering guides a pre-trained LLM through a system prompt that defines its role, tone and rules, optionally with a few example question–answer pairs. It requires no training and is the fastest approach to deploy, but the model can only use what is in the prompt and what it learned during pre-training.

2.2 Retrieval-Augmented Generation

RAG combines information retrieval with a language model. Documents such as FAQs, help articles and policies are split into chunks, embedded and indexed. At query time the retriever fetches the most relevant chunks and the model answers from them. Answer quality therefore depends strongly on the relevance and completeness of the retrieved evidence, and updating knowledge only requires re-indexing the documents.

2.3 Fine-tuning

Fine-tuning continues the training of a pre-trained model on domain examples, such as historical tickets and agent replies, so that behavior is built into the model weights. Parameter-efficient methods such as LoRA and QLoRA make this practical on modest hardware. Fine-tuning is well suited to consistent tone, output format and classification, but changing a policy requires preparing new data and retraining.

2.4 Hallucination and knowledge freshness

Hallucination is the generation of fluent but unsupported or incorrect content. In customer support it can lead to invented policies or promises the company cannot honor. Knowledge freshness is the ability of a system to reflect changed information, such as a new return window, without a costly rebuild. Both are central evaluation criteria for support automation.

3. Review of Related Research

Study 1: Fine-Tuning or Retrieval? Comparing Knowledge Injection in LLMs (Ovadia et al., 2024)

Approach: This study compares unsupervised fine-tuning with RAG as ways of adding knowledge to an LLM. Both are evaluated on knowledge-intensive question-answering tasks across several topics, covering facts the model saw during pre-training and entirely new facts.

Main findings: RAG consistently outperformed fine-tuning for both existing and new knowledge. The authors also report that LLMs struggle to learn new facts through fine-tuning, and that exposing the model to many variations of the same fact during training can reduce this problem.

Relevance and limitation: The work directly supports using RAG for fact-heavy support content. Its limitation for this project is that it studies general question answering with unsupervised fine-tuning, not support conversations, tone or hybrid configurations.

Source: https://arxiv.org/abs/2312.05934

Study 2: Does Fine-Tuning LLMs on New Knowledge Encourage Hallucinations? (Gekhman et al., 2024)

Approach: The authors design a controlled closed-book question-answering setup in which the share of fine-tuning examples that introduce new knowledge is varied, and they measure the effect on the model's use of its existing knowledge.

Main findings: Examples that introduce new knowledge are learned significantly more slowly than examples consistent with what the model already knows. Once such examples are eventually learned, they increase the model's tendency to hallucinate. The authors conclude that models mostly acquire factual knowledge in pre-training, while fine-tuning teaches them to use it more efficiently.

Relevance and limitation: This explains why fine-tuning is better suited to style, format and behavior than to storing company policies. The study is not set in customer support, so its findings should be re-tested on support data.

Source: https://aclanthology.org/2024.emnlp-main.444

Study 3: Retrieval-Augmented Generation with Knowledge Graphs for Customer Service Question Answering (Xu et al., 2024)

Approach: This work builds a knowledge graph from historical customer-service tickets so that retrieval preserves the structure within each ticket and the relations between tickets. At question time the system parses the query and retrieves related sub-graphs to generate the answer.

Main findings: The method is reported to improve retrieval and answer-generation metrics over a baseline, and after about six months of deployment at LinkedIn it reduced median issue resolution time by 28.6%.

Relevance and limitation: It shows that RAG works in a real customer-service deployment and that retrieval design strongly affects quality. It focuses on technical-support tickets and on improving RAG itself, and it does not compare RAG with prompting or fine-tuning.

Source: https://doi.org/10.1145/3626772.3661370

Study 4: Fine-Tuning or Fine-Failing? Debunking Performance Myths in Large Language Models (2024)

Approach: This study examines fine-tuning LLMs inside a RAG pipeline. The experiments were first run on private customer-inquiry data from a telecommunications company and then repeated on public datasets. It asks how fine-tuned models compare with their baseline counterparts in a RAG pipeline and whether training-set size affects the benefit.

Main findings: The study is directly about whether adding fine-tuning to RAG is worthwhile and how much data it needs. Its specific numerical results should be read from the full paper before they are cited.

Relevance and limitation: It is highly relevant to the hybrid configuration planned in this project. It concentrates on question answering within RAG and does not include a prompting-only baseline or a freshness test.

Source: https://arxiv.org/abs/2406.11201

Study 5: Enhancing Customer Support Chatbots with LLMs: Comparative Analysis of Few-Shot Learning, Fine-Tuning, and RAG (Woering, 2025)

Approach: This master's thesis compares few-shot learning, fine-tuning and RAG for customer-support chatbots in a first phase, and proposes an integrated architecture in a second phase.

Main findings: The thesis reports that RAG outperformed the other methods on all metrics, giving better response quality in customer interactions.

Relevance and limitation: It is the closest match to this project's topic. It is an unpublished thesis rather than a peer-reviewed paper, and its scope should be checked against the planned evaluation, which adds hallucination, latency, cost and policy-change adaptability.

Source: https://studenttheses.uu.nl/handle/20.500.12932/48393

Study 6: Adapting Large Language Models for Customer Service: A Comparative Study of RAG, RAFT and DoRA (2025)

Approach: This conference study compares Retrieval-Augmented Generation, Retrieval-Augmented Fine-Tuning (RAFT) and Weight-Decomposed Low-Rank Adaptation (DoRA) on a shared backbone model, using the Bitext customer-support dataset.

Main findings: RAG achieved strong factual consistency but with higher inference latency. The results highlight trade-offs among latency, accuracy and factual reliability, and the importance of data scale and retrieval grounding.

Relevance and limitation: It supports the use of the Bitext dataset and of latency as an evaluation dimension. It does not cover plain prompting and, as reported, does not examine how systems respond when policies change.

Source: https://5thworldconferenceoninformations.sched.com/event/2VAia/adapting-large-language-models-for-customer-service-a-comparative-study-of-rag-raft-and-dora

4. Comparison of Adaptation Techniques

Technique	How it works	Strength	Limitation
Prompting	Gives the model instructions, rules and a few examples in the prompt.	Fastest and cheapest to build; easy to iterate.	No company knowledge; may invent policies; long prompts raise cost.
RAG	Retrieves relevant chunks from a knowledge base and answers from them.	Grounded, can cite sources, updates when documents change.	Depends on retrieval quality; adds infrastructure and latency.
Fine-tuning	Trains the model further on past tickets and ideal replies.	Consistent tone and format; shorter prompts; good for classification.	Needs quality data; poor at adding facts; retraining needed for changes.
Hybrid (RAG + fine-tuned model)	Fine-tuning supplies style and format, RAG supplies current facts.	Combines grounding with consistent behavior.	Highest build and maintenance effort; gains must be verified.

5. Research Gap

Existing studies show that RAG is generally better than fine-tuning for supplying facts, that fine-tuning mainly shapes behavior, and that RAG can work in real customer-service settings. However, most of this work compares only two techniques, uses general question-answering tasks, or reports accuracy alone. There is room for a controlled comparison of prompting, RAG, fine-tuning and their hybrids on the same customer-support queries, using the same base model and test set. Such a study should measure accuracy, hallucination, tone, latency and cost per query, and should include a freshness test that changes a policy and checks how quickly each approach reflects it.

6. Proposed Research Direction

1. Select a customer-support dataset such as Bitext, or build a synthetic company knowledge base with FAQs and policy documents.
2. Create a test set covering FAQs, policy questions, order issues, emotional customers and out-of-scope questions.
3. Build a prompt-only baseline with zero-shot and few-shot variants.
4. Build a RAG pipeline with chunking, embeddings, a vector store and a retriever, and tune chunk size and top-k.
5. Fine-tune a smaller open model with LoRA or QLoRA on anonymized ticket data.
6. Build hybrid configurations, such as RAG with a fine-tuned model, with guardrails and human escalation.
7. Evaluate all systems on the same test set for correctness, faithfulness, tone, latency and cost, and run a policy-change experiment.

7. Conclusion

The reviewed literature indicates that RAG is the stronger technique for supplying accurate, current company knowledge, while fine-tuning is better suited to consistent behavior than to storing facts. For customer support, a promising direction is a hybrid design: RAG for facts, a prompted or fine-tuned model for tone and format, and guardrails with human escalation for safety. Its effectiveness must be validated experimentally rather than assumed.

References

1. Ovadia, O., Brief, M., Mishaeli, M., & Elisha, O. (2024). Fine-Tuning or Retrieval? Comparing Knowledge Injection in LLMs. Proceedings of EMNLP 2024. arXiv:2312.05934. https://arxiv.org/abs/2312.05934
2. Gekhman, Z., Yona, G., Aharoni, R., Eyal, M., Feder, A., Reichart, R., & Herzig, J. (2024). Does Fine-Tuning LLMs on New Knowledge Encourage Hallucinations? Proceedings of EMNLP 2024, 7765–7784. https://doi.org/10.18653/v1/2024.emnlp-main.444
3. Xu, Z., Cruz, M. J., Guevara, M., Wang, T., Deshpande, M., Wang, X., & Li, Z. (2024). Retrieval-Augmented Generation with Knowledge Graphs for Customer Service Question Answering. Proceedings of SIGIR 2024. https://doi.org/10.1145/3626772.3661370
4. Fine-Tuning or Fine-Failing? Debunking Performance Myths in Large Language Models. (2024). arXiv:2406.11201. https://arxiv.org/abs/2406.11201
5. Woering, R. (2025). Enhancing Customer Support Chatbots with LLMs: Comparative Analysis of Few-Shot Learning, Fine-Tuning, and RAG including the Proposal of an Integrated Architecture [Master's thesis]. Utrecht University. https://studenttheses.uu.nl/handle/20.500.12932/48393
6. Adapting Large Language Models for Customer Service: A Comparative Study of RAG, RAFT and DoRA. (2025). 5th World Conference on Informations. https://5thworldconferenceoninformations.sched.com/event/2VAia/adapting-large-language-models-for-customer-service-a-comparative-study-of-rag-raft-and-dora